

This test is a closed-book and closed-notes exam. No electric devices or calculators are allowed.

All of your solutions should be easily identifiable and supporting work must be shown. Please clearly label what work you want to be considered for grading. Ambiguous or illegible answers will not be counted as correct.

The duration of the exam is 50 minutes. If you finish early, feel free to submit your exam and leave the room, but not in the last 5 minutes.

Name: Exam Solutions

Student ID Number: _____

Signature: _____

I certify, on my honor, that I have neither given nor received any help, or used any non-permitted resources while completing this evaluation.

1. (25 points) **Fixed Point Damping:** The goal of this problem is to study the damped fixed point method with damping parameter $\lambda \in \mathbb{R}$ in $g(x) = x + \lambda f(x)$, where $f(x^*) = 0$ and $g(x^*) = x^*$.

- (a) (10 points) (i) First state the existence criteria for $g(x)$ to have a fixed point x^* on the interval $[a, b]$. (ii) Then state the criteria for there to exist a unique fixed point on $[a, b]$. (iii) Finally, state the key differences in the order of information required for uniqueness.

(i) If $g(x) \in C[a, b]$ and $g(x)$ maps $[a, b]$ into $[a, b]$ then $\exists x^* \in [a, b]$ s.t. $g(x^*) = x^*$.

(ii) If $g(x) \in C^1[a, b]$ and $g(x) \in [a, b] \forall x \in [a, b]$ and $\exists k \in \mathbb{R}^{>0}$ with $k < 1$ s.t. $|g'(x)| < k \forall x \in [a, b]$ then $\exists! x^*$ s.t. $g(x^*) = x^*$.

(iii) The first criteria only relies on zeroth-order information i.e. the continuity of g and its range. The second criteria relies on the first-order information of g namely its derivative.

- (b) (5 points) Let $f(x) = x^2 - 3.5x + 2.5$ with $f(1) = 0$ and consider the fixed point method with $g(x) = x + f(x)$ converging to $x^* = 1$. Determine whether the order of convergence is linear. (Hint: You can use a theorem from class to avoid deriving the limit.)

(In addition to the above)

If $g'(p) \neq 0$ and $|g'(p)| < 1$ then the method converges linearly.

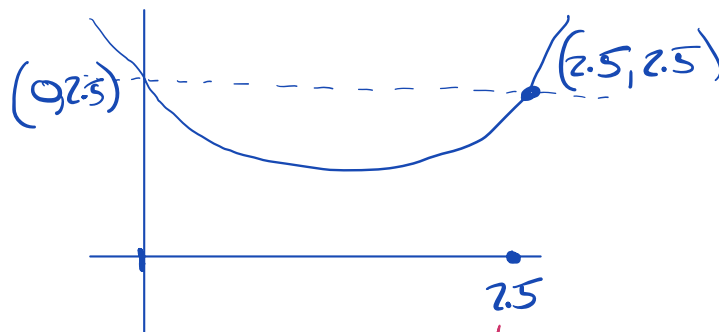
$$g(x) = x^2 - 2.5x + 2.5$$

$$g'(x) = 2x - 2.5$$

$$g'(1) = -0.5$$

$|g'(1)| < 1 \therefore$ the order of convergence is linear.

Note: On $[0, 2.5]$ $g(x)$ maps x into $[0, 2.5]$ satisfying the assumption



- (c) (10 points) Find the value of λ that achieves the highest order of convergence for $g_\lambda(x) = x + \lambda f(x)$. Justify your answer.
(Hint: Recall the second in class approach showing the order of convergence for Newton's Method.)

In this case let $g'_\lambda(p) = 0$ and find λ .

$$g'_\lambda(x) = 1 + f'(x) = 2(x - 3)^5 + 1 \quad @ \quad x = 1 \quad g'_\lambda(1) = -15\lambda + 1 = 0$$

then $\boxed{\lambda = \frac{2}{3}}$

This achieves second order convergence as demonstrated in class.

2. (25 points) **Convexity and Monotonicity and Newton's Method:** The goal of this problem is to study the effects of convexity and monotonicity on Newton's Method. We will focus on a function that is monotonically increasing where $f'(x) > 0$ and convex so that $f''(x) > 0$ for all $x \in [a, b]$.

- (a) (5 points) (i) Clearly state the criteria for the existence of a root of $f(x)$ on the interval $[a, b]$. (ii) Make a stronger assertion for the roots of $f(x)$ by assume that $f(x)$ is monotonically increasing on $[a, b]$. (You do not need to prove your statement.)

(i) If $f \in C[a, b]$ and $f(a)f(b) < 0$ then $\exists c \in [a, b]$
st $f(c) = 0$.

(ii) If $f \in C[a, b]$ with $f(a)f(b) < 0$ and $f'(x) > 0 \forall x \in [a, b]$
then $\exists! x^* \in [a, b]$ st $f(x^*) = 0$.

- (b) (5 points) Use pseudocode to define the Newton's Method routine. Make sure that you clearly define the Newton's step and any requirements on $f(x)$.

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Input  $x_0, N, \epsilon$ 
 $i = 1$ 
while  $i < N$ 
     $x = x_0 - \frac{f(x)}{f'(x)}$ 
    if  $|x - x_0| \geq \epsilon$ :
        return  $x_0$ 
     $i = i + 1$ 
     $x_0 = x$ 
return max iter reached.
```

- (c) (5 points) Let $f(x) = (x-1)^3 + (x-1)$ use Newton's method to compute x_1 and x_2 starting from $x_0 = 2$. (You do not need to simplify the expression for x_2 , but you should note if $x_2 \leq x^*$ or $x_2 \geq x^*$.)

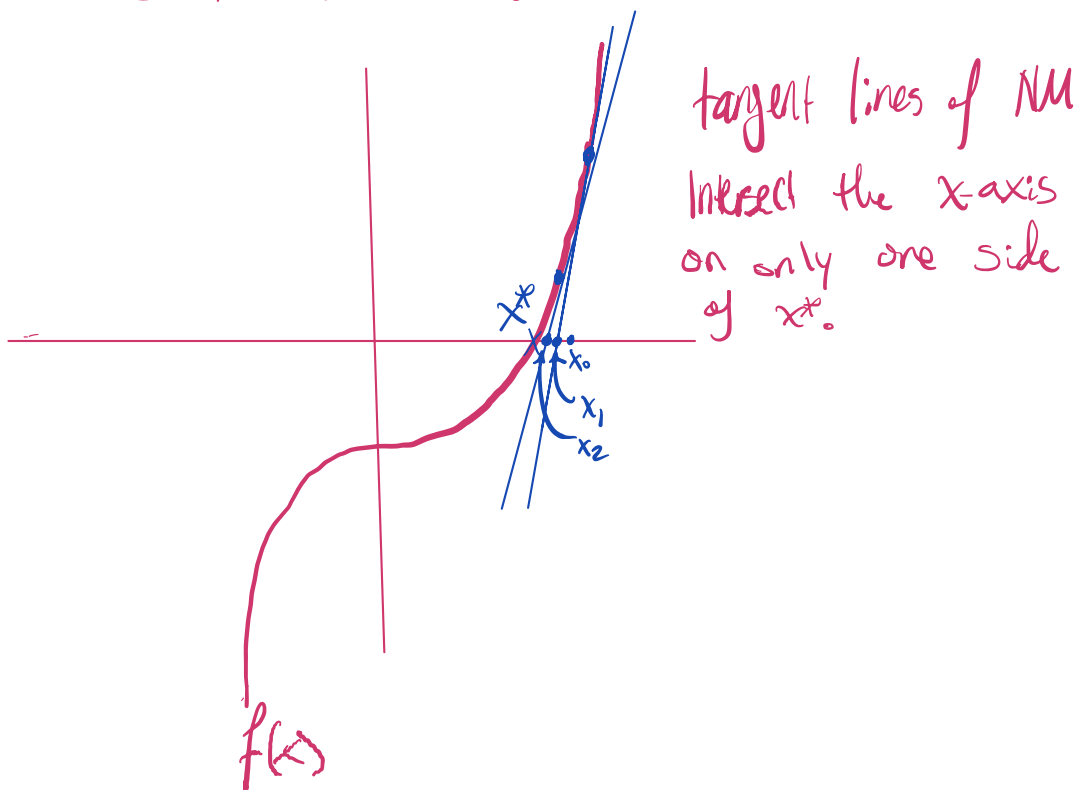
$$f'(x) = 3(x-1)^2 + 1 \quad x_0 = 2 \quad x_1 = 2 - \frac{1}{2} = 1.5$$

$$f(x_0) = 1^3 + 1 = 2 \quad f(x_1) = .5^3 + 1 \quad x_2 = 1.5 - \frac{.5^3 + 1}{3(.5)^2 + 1}$$

$$f'(x_0) = 3(1)^2 + 1 = 4 \quad f'(x_1) = 3(.5)^2 + 1$$

- (d) (10 points) Make a statement about the *bracketing* of Newton's method for convex and monotonically increasing functions (i.e., $x_0 \geq x_1 \geq x_2 \geq \dots \geq x^*$) that starts with an assumption on the starting side (i.e., $x_0 \geq x^*$ or $x_0 \leq x^*$). Use an illustration to justify your reasoning. (You do not need to prove your statement.)

Suppose $f \in C^2[a, b]$ with $f(a)f(b) < 0$, $f'(x) > 0$ and $f''(x) > 0 \quad \forall x \in [a, b]$ then $\exists! x^*$ s.t. $f(x^*) = 0$. Moreover if $x_0 > x^*$ then using Newton's Method $\forall i \in \mathbb{N} \quad x_i \geq x_{i+1} \geq x^*$.



Feel free to use this as scratch.